

Dr. Sudhir Kumar Sahu

CONTACT INFORMATION

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RESEARCH INTERESTS

circuit quantum electrodynamics (cQED), parametric amplification, qubit readout schemes, cavity electromechanics, and hybrid quantum systems.

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow

University of Colorado, Boulder, USA

In collaboration with National Institute of Standards and Technology (NIST)
July 2025 – Present

Postdoctoral Researcher

University of Massachusetts, Lowell, USA

In collaboration with National Institute of Standards and Technology (NIST)
Dec 2022 – June 2025

EDUCATION

Indian Institute of Science, Bangalore, India

Ph.D., Physics, 2016-2022

Thesis: “A cavity electromechanical device for superconducting vortex charge sensing”

Utkal University, Bhubaneswar, India

M.Sc., Condensed matter physics (Major), 2014-2016.

Sambalpur University, Patnagarh, India

B.Sc., Physics (Major), 2011-2014.

BRIEF SUMMARY OF RESEARCH

I am currently a postdoctoral fellow at the University of Colorado Boulder, conducting advanced research in collaboration with the National Institute of Standards and Technology (NIST), Boulder, Colorado, USA. My postdoctoral work focuses on expanding the capabilities of circuit quantum electrodynamics (cQED) architectures through parametric processes, such as beam-splitters, cross-Kerr, and two-photon squeezing. As my primary project, we developed a novel time-sequenced parametric protocol to achieve high-fidelity qubit readout. Here, we use the device's nonlinearity and two-photon driving to latch the memory mode onto one of two stable cavity states, depending on the qubit initialization, to achieve near-unity separation fidelity. In a collaborative effort, we have investigated fast, high-fidelity quantum logic operations by demonstrating an iSWAP gate controlled via a double-transmon coupler element. Furthermore, to mitigate performance-limiting energy loss within these superconducting quantum platforms, we fabricated and characterized low-loss Al/Si/Al parallel-plate capacitors utilizing a crystalline silicon (Si) dielectric configured in a FIN architecture.

During my Ph.D. at the Indian Institute of Science, Bangalore, my research centered on cavity electromechanics and quantum hybrid devices, culminating in my thesis titled "A cavity electromechanical device for superconducting vortex charge sensing". In my thesis work, I utilized a few unit cells of thin exfoliated crystals of the high- T_c superconductor (HTC) $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ as a mechanical resonator coupled to the fundamental mode of a superconducting coplanar waveguide cavity. With this electromechanical platform, we successfully measured different loss mechanisms in the mK temperature range and quantified elastic properties like Young's modulus and pre-stress using indentation and buckling verification techniques at room temperature. Ultimately, we achieved the ability to probe the effect of minuscule amounts of charge trapped within the superconducting vortex core of the HTS crystal. Besides my thesis project, I was continuously working on other collaborative work. For example, we have investigated chaotic behavior in an aluminum drum resonator operated in an ultra-strong coupling regime to a 3D aluminum cavity. In another work, we demonstrated magnetic flux-mediated coupling between an aluminum nanowire and a 3D microwave cavity mediated through a transmon qubit. Moreover, I contributed to the development of a Josephson Parametric Amplifier (JPA) for use as a quantum-noise-limited amplifier. In addition, I was involved in the development of a four-qubit quantum processor as part of the National Quantum Mission, an initiative of the Government of India.

PUBLICATIONS

1. A Time-Sequenced Parametric protocol for High-Fidelity latched Qubit readout in a Circuit-QED architecture.
Sudhir K. Sahu, Zachary L. Parrott, Trevyn F. Q. Larson, Anthony McFadden, Debsuvra Mukhopadhyay, Benton Miller, Kaixuan Ji, Florent Lecocq, Tarush Tiwari, Leonardo Ranzani, Archana Kamal and Raymond W. Simmonds.
[In preparation].
2. High-fidelity iSWAP gate with Double Transmon Coupler.

Tarush Tiwari, **Sudhir K. Sahu**, Guilhem Ribeill, Michael Senatore, Matthew D. La-Haye, Raymond W. Simmonds, Daniel L. Campbell, Archana Kamal, Leonardo Ranzani [arXiv:2604.27080 \[quant-ph\]](#).

3. Fabrication and characterization of low-loss Al/Si/Al parallel plate capacitors for superconducting quantum information applications.
Anthony McFadden, Aranya Goswami, Tongyu Zhao, Teun van Schijndel, Trevyn F.Q. Larson, **Sudhir kumar Sahu**, Stephen Gill, Florent Lecocq, Raymond Simmonds, Chris Palmstrøm.
[npj Quantum Inf 11, 11 \(2025\)](#).
4. Instabilities near Ultrastrong Coupling in a Microwave Optomechanical Cavity.
Soumya Ranjan Das, Sourav Majumder, **Sudhir Kumar Sahu**, Ujjawal Singhal, Tanmoy Bera, and Vibhor Singh.
[Phys. Rev. Lett. 131, 067001 \(2023\)](#).
5. Superconducting vortex-charge measurement using cavity electromechanics.
Sudhir Kumar Sahu, Supriya Mandal, Sanat Ghosh, Mandar M. Deshmukh, and Vibhor Singh.
[Nano Lett. 22, 4, 1665 \(2022\)](#).
6. Large flux-mediated coupling in hybrid electromechanical system with a transmon qubit.
Tanmoy Bera, Sourav Majumder, **Sudhir Kumar Sahu**, and Vibhor Singh.
[Communications Physics. 4, 1-7\(2021\)](#).
7. Nanoelectromechanical resonators from high- T_c superconducting crystals of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$.
Sudhir Kumar Sahu, Jaykumar Vaidya, Felix Schmidt, Digambar Jangade, Arumugam Thamizhavel, Gary Steele, Mandar M. Deshmukh, and Vibhor Singh.
[2D Mater. 6, 025027\(2019\)](#).
8. Elastic properties of few unit cell thick superconducting crystals of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$.
Sudhir Kumar Sahu, Digambar Jangade, Arumugam Thamizhavel, Mandar M. Deshmukh, and Vibhor Singh.
[Appl. Phys. Lett. 115, 143102 \(2019\)](#).
9. Cavity Optomechanical Platform with a Three-dimensional Waveguide Cavity.
Bindu Gunupudi, Soumya Ranjan Das, Rohit Navarathna, **Sudhir Kumar Sahu**, Sourav Majumder, and Vibhor Singh.
[Phys. Rev. Applied. 11, 024067\(2019\)](#).

SCIENTIFIC TALKS

- APS Global Summit 2026, Denver, Colorado (USA).
(Title: High-fidelity pulsed qubit readout with a latched cavity state in a parametric circuit-QED system.)
- APS Global Summit 2025, Anaheim, California (USA).
(Title: High fidelity and efficient qubit measurement using parametric protocols in a circuit-QED setup.)
- APS March Meeting 2024, Minneapolis, Minnesota (USA).
(Title: High-fidelity qubit readout with parametric protocols and on-chip amplification in a circuit-QED setup.)
- APS March Meeting 2021, virtual mode.
(Title: Observation of magnetic hysteresis and vortex charge in a radiation pressure force driven superconducting $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ micromechanical resonator.)
- Student seminar series at Department of Physics, IISc, Bangalore, India, Sept 2019.
(Title: Cavity-optomechanical devices based on exfoliable superconducting crystals of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$.)

POSTER PRESENTATIONS

- “In-house Symposium” at Department of Physics, IISc, Bangalore, India, Jan 2022.
- Quantum Materials (Qmat) July 2019, Bangalore, India.
(Title: Cavity-optomechanical devices based on exfoliable superconducting crystals of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$.)
- APS March Meeting 2019, Boston, USA.
(Title: Nanoelectromechanical resonators from thin superconducting crystals of $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$.)
- “In-house Symposium” at Department of Physics, IISc, Bangalore, India, November 2019.
- “In-house Symposium” at Department of Physics, IISc, Bangalore, India, November 2018.

TEACHING EXPERIENCE

- Teaching Assistant (Jan–Apr 2022)
Course: PH 207: Analog digital and microprocessor electronics-1
- Teaching Assistant (July–Dec 2021)
Course: QT 201: Quantum technology lab
- Teaching Assistant (July–Dec 2018)
Course: UP 201: Introductory physics III – thermal and modern physics

SCHOLARSHIPS & AWARDS

- UGC JRF JUNE 2015, All India Rank-118
- GATE 2016, SCORE 738/1000, All India Rank-38

SKILLS & ABILITIES

- **Programming Languages:** Python, Fortran, Qiskit Metal, QuTiP, etc.
- **Applications:** Adobe Illustrator, LayoutEditor, AutoCAD, Microsoft Office, L^AT_EX, etc.
- **Simulation Software:** COMSOL, Mathematica, MATLAB, Microwave Office AWR, ANSYS.
- **Nanofabrication & Characterization:** E-beam lithography, photolithography, reactive ion etching, critical point dryer, Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), Dektak, e-beam evaporator, sputtering, Chemical Vapor Deposition (CVD), thermal evaporation.
- **Packaging:** Wire bonder, soldering station, etc.
- **Microwave Measurements:** Arbitrary Signal Generator with FPGA (Qblox, Quantum Machines), Vector Network Analyzer (VNA), spectrum analyzer, oscilloscope, lock-in amplifier, arbitrary waveform generator, cryogenic probe station, etc.
- **Low Temperature & Vacuum Systems:** Oxford dilution refrigerator, Bluefors dilution refrigerator, leak detector, cryo pumps, dry vacuum pumps.

ACADEMIC
REFERENCES

Prof. Raymond W. Simmonds

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Advanced Microwave Photonics Group
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Boulder, CO 80305
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